



Tesla in-car karaoke

TeslaMic, Tesla's \$188 in-car karaoke microphone, was sold out within an hour of going on sale in China <https://digit.in/mar22-05>



Meta 3-D Avatars

Users can now share 3-D avatars on Instagram stories and direct messages <https://digit.in/mar22-06>



CRYPTOCURRENCY AND CYBERSECURITY: A COMPLICATED RELATIONSHIP?

Do you think your Cryptocurrency investment is really secure? Read on to find out!

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A

relatively lesser known name in the decentralised finance space, Poly Network was suddenly forced

into the limelight for all the wrong reasons on 8th August 2021, when it announced that it had witnessed what was being referred to as the largest cryptocurrency theft in history. In a single heist, hackers stole \$613 million in digital coins from the interoperability protocol for heterogeneous blockchains. In simpler terms, a blockchain bridge, which allows for users to swap tokens, or digital coins in this case, across different platforms.

In a similar incident, hackers stole from the Wormhole Portal which allows easy swapping of cryptocurren-

cies and non-fungible tokens between the Solana and Ethereum blockchains, and stole 120,000 "Wrapped Ethereum" tokens valued at approximately \$320 million in February 2022.

The aforementioned thefts are just two of a multitude of similar incidents raising an extremely pertinent question – is cryptocurrency, and its underlying technology, blockchain, really as secure as claimed?

HOW EXACTLY DOES BLOCKCHAIN TECHNOLOGY WORK?

A good starting point to answer that would be to understand the technology itself. And what better example than Bitcoin? In Bitcoin's blockchain, the history of every Bitcoin transaction ever made is stored as an accounting ledger. Multiple copies of this ledger are stored on a network of computers referred to as nodes. Each time someone makes a transaction, these nodes check its validity – that is, to verify that whoever spent a Bitcoin actually had one to spend. A subset of these nodes package these transactions into the ledger as "blocks". The owners of these nodes that do so are referred to as miners, and

earn Bitcoins as rewards for adding new blocks to the chain. The system, on paper at least – is foolproof owing to the presence of two elements. The first is the Cryptographic Fingerprint, also referred to as a hash. Each hash takes a lot of computing time to generate, and works to serve as proof that the miner who added a block to the Bitcoin blockchain did the computational work required to do so. This is also referred to as a Proof-of-Work protocol. Once nodes have verified that a hash indeed belongs to the block it shows up in, they update their copies of the ledger with the new block.

There is also a third element which further secures the system. Each new block includes the previous block's unique hash. To alter a block in the ledger, one has to calculate a new hash not just for the block it is present in, but also every new block. Unless a miner is fast enough, which they can be provided they have computers faster than the rest of the nodes in the system combined, any block they add will conflict with existing ones, and nodes will reject these alterations.

COMMON METHODS USED FOR CRYPTOCURRENCY THEFT

a) Phishing:

Phishing campaigns refer to targeting user credentials on cryptocurrency trading platforms. The most common way hackers achieve the same is by using homograph domains. In essence, it refers to a malicious party, tricking computer users into thinking that they are communicating with a legitimate domain by creating fake websites which pose as original ones and disclosing their user credentials. The way hackers get users to access these spoofed websites is by sending them emails with links to the fake websites.

b) Compromised Registration Forms:

Most cryptocurrency platforms have an extremely strict user identity verification process. However, hackers often directly inject codes into these registration forms, which when filled, send the